Substitution and Trigonometric Integrals— Solutions

Warm-up

Question 1a:
To evaluate $\int \frac{2x}{1+x^2} dx$, we use the substitution method.
Let $u = 1 + x^2$.
Then, differentiating with respect to x:

$$\begin{aligned} \frac{du}{dx} &= 2x \\ du &= 2x\,dx \end{aligned}$$

Substituting these into the integral:

$$\begin{aligned} \int \frac{2x}{1+x^2} dx &= \int \frac{1}{u}\,du \\ &= \ln|u| + C \\ &= \ln(1+x^2) + C \end{aligned}$$

Since $1 + x^2 > 0$ for all real x, the absolute value is not required.

Question 1b:
We are given that $\int \frac{1}{1+x^2} dx = \tan^{-1} x + C$ and from part (a), $\int \frac{2x}{1+x^2} dx = \ln(1+x^2) + C$.
First, we rewrite the integral as a sum of two fractions:

$$\begin{aligned} \int \frac{x+1}{1+x^2} dx &= \int \left(\frac{x}{1+x^2} + \frac{1}{1+x^2} \right) dx \\ &= \int \frac{x}{1+x^2} dx + \int \frac{1}{1+x^2} dx \end{aligned}$$

From part (a), we know $\int \frac{2x}{1+x^2} dx = \ln(1+x^2)$, so $\int \frac{x}{1+x^2} dx = \frac{1}{2}\ln(1+x^2)$.
Applying the known integrals:

$$\int \frac{x+1}{1+x^2} dx = \frac{1}{2}\ln(1+x^2) + \tan^{-1} x + C$$

Question 2:
To evaluate $\int \frac{x}{x+1} dx$, we use algebraic manipulation to simplify the integrand.
Add and subtract 1 in the numerator:

$$\begin{aligned} \int \frac{x}{x+1} dx &= \int \frac{x+1-1}{x+1} dx \\ &= \int \left(\frac{x+1}{x+1} - \frac{1}{x+1} \right) dx \\ &= \int (1 - \frac{1}{x+1}) dx \end{aligned}$$

Integrating term by term:

$$\int (1 - \frac{1}{x+1}) dx = x - \ln|x+1| + C$$

Question 3:
To evaluate $\int \frac{x^2}{x^2+x+1}dx$, we perform polynomial long division since the degree of the numerator is equal to the degree of the denominator.

Dividing x^2 by x^2+x+1:

$$\begin{aligned}\frac{x^2}{x^2+x+1} &= \frac{(x^2+x+1)-(x+1)}{x^2+x+1} \\ &= 1 - \frac{x+1}{x^2+x+1}\end{aligned}$$

Now we integrate:

$$\begin{aligned}\int \frac{x^2}{x^2+x+1}dx &= \int 1dx - \int \frac{x+1}{x^2+x+1}dx \\ &= x - \int \frac{x+1}{x^2+x+1}dx\end{aligned}$$

For the second integral, we manipulate the numerator to match the derivative of the denominator $(2x+1)$:

$$\begin{aligned}\int \frac{x+1}{x^2+x+1}dx &= \int \frac{\frac{1}{2}(2x+1)+\frac{1}{2}}{x^2+x+1}dx \\ &= \frac{1}{2}\int \frac{2x+1}{x^2+x+1}dx + \frac{1}{2}\int \frac{1}{x^2+x+1}dx\end{aligned}$$

The first part is a natural log form, and the second part requires completing the square:

$$\int \frac{2x+1}{x^2+x+1}dx = \ln|x^2+x+1|$$
$$\int \frac{1}{x^2+x+1}dx = \int \frac{1}{(x+\frac{1}{2})^2+\frac{3}{4}}dx = \frac{1}{\sqrt{3}/2}\tan^{-1}\left(\frac{x+1/2}{\sqrt{3}/2}\right) = \frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)$$

Combining everything:

$$\int \frac{x^2}{x^2+x+1}dx = x - \frac{1}{2}\ln|x^2+x+1| - \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

Question 4:
To evaluate $\int_0^{\sqrt{3}} \frac{x+2}{\sqrt{x^2+1}}dx$, we split the integral into two parts:

$$\int_0^{\sqrt{3}} \frac{x+2}{\sqrt{x^2+1}}dx = \int_0^{\sqrt{3}} \frac{x}{\sqrt{x^2+1}}dx + \int_0^{\sqrt{3}} \frac{2}{\sqrt{x^2+1}}dx$$

For the first integral, let $u = x^2+1$, then $du = 2xdx$, so $xdx = \frac{1}{2}du$. Limits: $x = 0 \implies u = 1$; $x = \sqrt{3} \implies u = 4$.

$$\begin{aligned}\int_0^{\sqrt{3}} \frac{x}{\sqrt{x^2+1}}dx &= \frac{1}{2}\int_1^4 u^{-1/2}du \\ &= \frac{1}{2}[2u^{1/2}]_1^4 \\ &= [\sqrt{u}]_1^4 = 2 - 1 = 1\end{aligned}$$

For the second integral, we use the standard formula $\int \frac{1}{\sqrt{x^2+a^2}}dx = \ln|x + \sqrt{x^2 + a^2}| + C$.

$$\begin{aligned}
\int_0^{\sqrt{3}} \frac{2}{\sqrt{x^2+1}}dx &= 2[\ln|x + \sqrt{x^2+1}|]_0^{\sqrt{3}} \\
&= 2(\ln|\sqrt{3} + \sqrt{3+1}| - \ln|0 + \sqrt{0+1}|) \\
&= 2(\ln|\sqrt{3} + 2| - \ln 1) \\
&= 2\ln(2 + \sqrt{3})
\end{aligned}$$

Summing the results:

$$\int_0^{\sqrt{3}} \frac{x+2}{\sqrt{x^2+1}}dx = 1 + 2\ln(2 + \sqrt{3})$$

Question 5:
To evaluate $\int \frac{x}{(1+x^2)^3}dx$, we use substitution.
Let $u = 1 + x^2$, then $du = 2xdx$, so $xdx = \frac{1}{2}du$.

$$\begin{aligned}
\int \frac{x}{(1+x^2)^3}dx &= \int \frac{1}{u^3} \cdot \frac{1}{2}du \\
&= \frac{1}{2}\int u^{-3}du \\
&= \frac{1}{2}\left(\frac{u^{-2}}{-2}\right) + C \\
&= -\frac{1}{4u^2} + C
\end{aligned}$$

Substituting back $u = 1 + x^2$:

$$\int \frac{x}{(1+x^2)^3}dx = -\frac{1}{4(1+x^2)^2} + C$$

Question 6:
To evaluate $\int e^x \cos(e^x)dx$, we use substitution.
Let $u = e^x$, then $du = e^x dx$.

$$\begin{aligned}
\int e^x \cos(e^x)dx &= \int \cos(u)du \\
&= \sin(u) + C
\end{aligned}$$

Substituting back $u = e^x$:

$$\int e^x \cos(e^x)dx = \sin(e^x) + C$$

Question 7:
To evaluate $\int \frac{1}{\sqrt{x}}e^{\sqrt{x}}dx$, we use substitution.
Let $u = \sqrt{x}$, then $du = \frac{1}{2\sqrt{x}}dx$, so $\frac{1}{\sqrt{x}}dx = 2du$.

$$\begin{aligned}
\int \frac{1}{\sqrt{x}} e^{\sqrt{x}} dx &= \int e^u (2du) \\
&= 2\int e^u du \\
&= 2e^u + C
\end{aligned}$$

Substituting back $u = \sqrt{x}$:

$$\int \frac{1}{\sqrt{x}} e^{\sqrt{x}} dx = 2e^{\sqrt{x}} + C$$

Question 8:
To evaluate $3\int \sqrt{9-x^2}dx$, we use the trigonometric substitution $x = 3\sin\theta$.
Then $dx = 3\cos\theta d\theta$ and $\sqrt{9-x^2} = \sqrt{9-9\sin^2\theta} = 3\cos\theta$.

$$\begin{aligned}
3\int \sqrt{9-x^2}dx &= 3\int (3\cos\theta)(3\cos\theta d\theta) \\
&= 3\int 9\cos^2\theta d\theta \\
&= 27\int \frac{1+\cos 2\theta}{2} d\theta \\
&= \frac{27}{2}\int (1+\cos 2\theta) d\theta \\
&= \frac{27}{2}\left(\theta + \frac{1}{2}\sin 2\theta\right) + C
\end{aligned}$$

Using $\sin 2\theta = 2\sin\theta\cos\theta$:

$$\frac{27}{2}(\theta + \sin\theta\cos\theta) + C$$

From $x = 3\sin\theta$, we have $\sin\theta = x/3$ and $\theta = \sin^{-1}(x/3)$. Also $\cos\theta = \sqrt{1-(x/3)^2} = \frac{\sqrt{9-x^2}}{3}$.

$$\begin{aligned}
\text{Result} &= \frac{27}{2}\left(\sin^{-1}(x/3) + \frac{x}{3}\cdot\frac{\sqrt{9-x^2}}{3}\right) + C \\
&= \frac{27}{2}\sin^{-1}(x/3) + \frac{3x\sqrt{9-x^2}}{2} + C
\end{aligned}$$

Question 9:
To evaluate $\int \frac{1}{x^2\sqrt{x^2-4}}dx$, we use the substitution $x = 2\sec\theta$.
Then $dx = 2\sec\theta\tan\theta d\theta$ and $\sqrt{x^2-4} = \sqrt{4\sec^2\theta - 4} = 2\tan\theta$.

$$\begin{aligned}
\int \frac{1}{x^2\sqrt{x^2-4}}dx &= \int \frac{2\sec\theta\tan\theta d\theta}{(2\sec\theta)^2(2\tan\theta)} \\
&= \int \frac{2\sec\theta\tan\theta}{4\sec^2\theta\cdot 2\tan\theta} d\theta \\
&= \int \frac{1}{4\sec\theta} d\theta \\
&= \frac{1}{4}\int \cos\theta d\theta \\
&= \frac{1}{4}\sin\theta + C
\end{aligned}$$

From $x = 2\sec\theta$, we have $\cos\theta = 2/x$. Thus $\sin\theta = \sqrt{1-(2/x)^2} = \frac{\sqrt{x^2-4}}{x}$.

$$\int \frac{1}{x^2\sqrt{x^2-4}}dx = \frac{\sqrt{x^2-4}}{4x} + C$$

Question 10:
To evaluate $\int_0^{\pi/2} \frac{1}{1+\sin\theta}d\theta$, we use the Weierstrass substitution $x = \tan(\theta/2)$.
Then $\sin\theta = \frac{2x}{1+x^2}$, $d\theta = \frac{2dx}{1+x^2}$. Limits: $\theta = 0 \implies x = 0$; $\theta = \pi/2 \implies x = 1$.

$$\begin{aligned}\int_0^{\pi/2} \frac{1}{1+\sin\theta}d\theta &= \int_0^1 \frac{1}{1+\frac{2x}{1+x^2}} \cdot \frac{2dx}{1+x^2} \\ &= \int_0^1 \frac{1+x^2}{1+x^2+2x} \cdot \frac{2dx}{1+x^2} \\ &= \int_0^1 \frac{2}{(x+1)^2}dx\end{aligned}$$

Integrating:

$$\begin{aligned}\int_0^1 2(x+1)^{-2}dx &= [-2(x+1)^{-1}]_0^1 \\ &= [-\frac{2}{x+1}]_0^1 \\ &= -\frac{2}{2} - (-\frac{2}{1}) = -1 + 2 = 1\end{aligned}$$

Question 11:
To evaluate $\int \frac{1}{\sin x + \cos x}dx$, we use the substitution $u = \tan(x/2)$.
Then $\sin x = \frac{2u}{1+u^2}$, $\cos x = \frac{1-u^2}{1+u^2}$, and $dx = \frac{2du}{1+u^2}$.

$$\begin{aligned}\int \frac{1}{\sin x + \cos x}dx &= \int \frac{1}{\frac{2u}{1+u^2} + \frac{1-u^2}{1+u^2}} \cdot \frac{2du}{1+u^2} \\ &= \int \frac{1+u^2}{2u+1-u^2} \cdot \frac{2du}{1+u^2} \\ &= \int \frac{2}{1+2u-u^2}du\end{aligned}$$

Complete the square for the denominator:

$$\begin{aligned}1+2u-u^2 &= -(u^2-2u-1) \\ &= -((u-1)^2-2) = 2-(u-1)^2\end{aligned}$$

Now the integral is in the form $\int \frac{1}{a^2-u^2}du = \frac{1}{2a}\ln\left|\frac{a+u}{a-u}\right| + C$.

$$\begin{aligned}\int \frac{2}{2-(u-1)^2}du &= 2\int \frac{1}{(\sqrt{2})^2-(u-1)^2}du \\ &= 2 \cdot \frac{1}{2\sqrt{2}} \ln\left|\frac{\sqrt{2}+(u-1)}{\sqrt{2}-(u-1)}\right| + C \\ &= \frac{1}{\sqrt{2}} \ln\left|\frac{\sqrt{2}+u-1}{\sqrt{2}-u+1}\right| + C\end{aligned}$$

Substituting back $u = \tan(x/2)$:

$$\int \frac{1}{\sin x + \cos x} dx = \frac{1}{\sqrt{2}} \ln \left| \frac{\sqrt{2} + \tan(x/2) - 1}{\sqrt{2} - \tan(x/2) + 1} \right| + C$$

Skill-building

Question 1:
To evaluate $\int_1^3 \frac{\sqrt{x}}{x+1} dx$, we use the substitution $u = \sqrt{x}$.

Then $x = u^2$ and $dx = 2udu$. Limits: $x = 1 \implies u = 1$; $x = 3 \implies u = \sqrt{3}$.

$$\begin{aligned}
\int_1^3 \frac{\sqrt{x}}{x+1} dx &= \int_1^{\sqrt{3}} \frac{u}{u^2+1} (2udu) \\
&= 2 \int_1^{\sqrt{3}} \frac{u^2}{u^2+1} du \\
&= 2 \int_1^{\sqrt{3}} \frac{(u^2+1) - 1}{u^2+1} du \\
&= 2 \int_1^{\sqrt{3}} (1 - \frac{1}{u^2+1}) du
\end{aligned}$$

Integrating:

$$\begin{aligned}
&= 2[u - \tan^{-1} u]_1^{\sqrt{3}} \\
&= 2((\sqrt{3} - \tan^{-1}\sqrt{3}) - (1 - \tan^{-1} 1)) \\
&= 2(\sqrt{3} - \frac{\pi}{3} - 1 + \frac{\pi}{4}) \\
&= 2(\sqrt{3} - 1 - \frac{\pi}{12}) \\
&= 2\sqrt{3} - 2 - \frac{\pi}{6}
\end{aligned}$$

Question 2:
To evaluate $\int_{1/2}^3 \frac{x-1}{\sqrt{2x+3}} dx$, we use the substitution $u = 2x + 3$.

Then $x = \frac{u-3}{2}$ and $dx = \frac{1}{2} du$. Limits: $x = 1/2 \implies u = 4$; $x = 3 \implies u = 9$.

$$\begin{aligned}
\int_{1/2}^3 \frac{x-1}{\sqrt{2x+3}} dx &= \int_4^9 \frac{\frac{u-3}{2} - 1}{\sqrt{u}} \cdot \frac{1}{2} du \\
&= \frac{1}{4} \int_4^9 \frac{u - 3 - 2}{\sqrt{u}} du \\
&= \frac{1}{4} \int_4^9 \frac{u-5}{\sqrt{u}} du \\
&= \frac{1}{4} \int_4^9 (u^{1/2} - 5u^{-1/2}) du
\end{aligned}$$

Integrating:

$$= \frac{1}{4}[\frac{2}{3}u^{3/2} - 10u^{1/2}]_4^9$$
$$= \frac{1}{4}((\frac{2}{3}(27) - 10(3)) - (\frac{2}{3}(8) - 10(2)))$$
$$= \frac{1}{4}((18 - 30) - (\frac{16}{3} - 20))$$
$$= \frac{1}{4}(-12 - \frac{16}{3} + 20)$$
$$= \frac{1}{4}(8 - \frac{16}{3}) = \frac{1}{4}(\frac{24 - 16}{3}) = \frac{1}{4} \cdot \frac{8}{3} = \frac{2}{3}$$

Question 3:
To evaluate $\int \frac{1}{e^x+1} dx$, we can use a substitution or a trick.
Method: Add and subtract e^x in the numerator.

$$\int \frac{1}{e^x + 1} dx = \int \frac{1 + e^x - e^x}{e^x + 1} dx$$
$$= \int (\frac{1 + e^x}{e^x + 1} - \frac{e^x}{e^x + 1}) dx$$
$$= \int (1 - \frac{e^x}{e^x + 1}) dx$$

Integrating term by term:

$$\int 1 dx - \int \frac{e^x}{e^x + 1} dx = x - \ln|e^x + 1| + C$$

Question 4:
To evaluate $\int_0^1 \frac{x}{x^4+1} dx$, we use the substitution $u = x^2$.
Then $du = 2x dx$, so $x dx = \frac{1}{2} du$. Limits: $x = 0 \implies u = 0$; $x = 1 \implies u = 1$.

$$\int_0^1 \frac{x}{x^4 + 1} dx = \frac{1}{2} \int_0^1 \frac{1}{u^2 + 1} du$$
$$= \frac{1}{2}[\tan^{-1} u]_0^1$$
$$= \frac{1}{2}(\tan^{-1} 1 - \tan^{-1} 0)$$
$$= \frac{1}{2}(\frac{\pi}{4} - 0) = \frac{\pi}{8}$$

Question 5:
To evaluate $\int \frac{e^x+e^{2x}}{1+e^{2x}} dx$, we split the integral into two parts:

$$\int \frac{e^x + e^{2x}}{1 + e^{2x}} dx = \int \frac{e^x}{1 + e^{2x}} dx + \int \frac{e^{2x}}{1 + e^{2x}} dx$$

For the first integral, let $u = e^x$, then $du = e^x dx$.

$$\int \frac{e^x}{1 + e^{2x}} dx = \int \frac{1}{1 + u^2} du = \tan^{-1} u + C = \tan^{-1}(e^x) + C$$

For the second integral, let $v = 1 + e^{2x}$, then $dv = 2e^{2x}dx$, so $e^{2x}dx = \frac{1}{2}dv$.

$$\int \frac{e^{2x}}{1 + e^{2x}}dx = \frac{1}{2}\int \frac{1}{v}dv = \frac{1}{2}\ln|v| + C = \frac{1}{2}\ln(1 + e^{2x}) + C$$

Combining the results:

$$\int \frac{e^x + e^{2x}}{1 + e^{2x}}dx = \tan^{-1}(e^x) + \frac{1}{2}\ln(1 + e^{2x}) + C$$

Question 6:
To evaluate $\int x\sqrt{x + 1}dx$, we use the substitution $u = x + 1$.
Then $x = u - 1$ and $dx = du$.

$$\begin{aligned}
\int x\sqrt{x + 1}dx &= \int (u - 1)\sqrt{u}du \\
&= \int (u^{3/2} - u^{1/2})du \\
&= \frac{u^{5/2}}{5/2} - \frac{u^{3/2}}{3/2} + C \\
&= \frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2} + C
\end{aligned}$$

Substituting back $u = x + 1$:

$$\int x\sqrt{x + 1}dx = \frac{2}{5}(x + 1)^{5/2} - \frac{2}{3}(x + 1)^{3/2} + C$$

Question 7:
To evaluate $\int \frac{\sqrt{1-x^2}}{x}dx$, we use the substitution $x = \sin\theta$.
Then $dx = \cos\theta d\theta$ and $\sqrt{1 - x^2} = \cos\theta$.

$$\begin{aligned}
\int \frac{\sqrt{1 - x^2}}{x}dx &= \int \frac{\cos\theta}{\sin\theta}\cos\theta d\theta \\
&= \int \frac{\cos^2\theta}{\sin\theta}d\theta \\
&= \int \frac{1 - \sin^2\theta}{\sin\theta}d\theta \\
&= \int (\csc\theta - \sin\theta)d\theta
\end{aligned}$$

Integrating:

$$\begin{aligned}
&= -\ln|\csc\theta + \cot\theta| - (-\cos\theta) + C \\
&= -\ln|\csc\theta + \cot\theta| + \cos\theta + C
\end{aligned}$$

From $x = \sin\theta$, we have $\csc\theta = 1/x$ and $\cot\theta = \frac{\sqrt{1-x^2}}{x}$. Also $\cos\theta = \sqrt{1 - x^2}$.

$$\int \frac{\sqrt{1 - x^2}}{x}dx = -\ln\left|\frac{1 + \sqrt{1 - x^2}}{x}\right| + \sqrt{1 - x^2} + C$$

Question 8:
To evaluate $\int \sqrt{1-x^2}dx$, we use the substitution $x = \sin\theta$.
Then $dx = \cos\theta d\theta$ and $\sqrt{1-x^2} = \cos\theta$.

$$\begin{aligned}\int \sqrt{1-x^2}dx &= \int \cos\theta \cdot \cos\theta d\theta \\ &= \int \cos^2\theta d\theta \\ &= \int \frac{1+\cos 2\theta}{2} d\theta \\ &= \frac{1}{2}(\theta + \frac{1}{2}\sin 2\theta) + C \\ &= \frac{1}{2}(\theta + \sin\theta\cos\theta) + C\end{aligned}$$

From $x = \sin\theta$, we have $\theta = \sin^{-1} x$ and $\cos\theta = \sqrt{1-x^2}$.

$$\int \sqrt{1-x^2}dx = \frac{1}{2}(\sin^{-1} x + x\sqrt{1-x^2}) + C$$

Question 9:
To evaluate $\int \frac{\sqrt{x}}{1+x}dx$, we use the substitution $u = \sqrt{x}$.
Then $x = u^2$ and $dx = 2udu$.

$$\begin{aligned}\int \frac{\sqrt{x}}{1+x}dx &= \int \frac{u}{1+u^2}(2udu) \\ &= 2\int \frac{u^2}{1+u^2}du \\ &= 2\int \frac{(u^2+1)-1}{1+u^2}du \\ &= 2\int (1 - \frac{1}{1+u^2})du \\ &= 2(u - \tan^{-1} u) + C\end{aligned}$$

Substituting back $u = \sqrt{x}$:

$$\int \frac{\sqrt{x}}{1+x}dx = 2\sqrt{x} - 2\tan^{-1}\sqrt{x} + C$$

Question 10:
To evaluate $\int \frac{1}{\sqrt{a^2+x^2}}dx$, we use the substitution $x = a\tan\theta$.
Then $dx = a\sec^2\theta d\theta$ and $\sqrt{a^2+x^2} = a\sec\theta$.

$$\begin{aligned}\int \frac{1}{\sqrt{a^2+x^2}}dx &= \int \frac{a\sec^2\theta d\theta}{a\sec\theta} \\ &= \int \sec\theta d\theta \\ &= \ln|\sec\theta + \tan\theta| + C\end{aligned}$$

From $x = a\tan\theta$, we have $\tan\theta = x/a$ and $\sec\theta = \frac{\sqrt{a^2+x^2}}{a}$.

$$
\begin{aligned}
\int \frac{1}{\sqrt{a^2+x^2}}dx &= \ln\left|\frac{\sqrt{a^2+x^2}}{a} + \frac{x}{a}\right| + C \\
&= \ln|\sqrt{a^2+x^2} + x| - \ln a + C \\
&= \ln|x + \sqrt{a^2+x^2}| + C'
\end{aligned}
$$

Question 11:

To evaluate $\int \csc x dx$, we use the substitution $x = \tan(\theta/2)$.

Then $\sin x = \frac{2\tan(\theta/2)}{1+\tan^2(\theta/2)} = \frac{2u}{1+u^2}$ where $u = \tan(\theta/2)$. Also $dx = \frac{2du}{1+u^2}$.

$$
\begin{aligned}
\int \csc x dx = \int \frac{1}{\sin x}dx &= \int \frac{1}{\frac{2u}{1+u^2}} \cdot \frac{2du}{1+u^2} \\
&= \int \frac{1+u^2}{2u} \cdot \frac{2du}{1+u^2} \\
&= \int \frac{1}{u}du \\
&= \ln|u| + C
\end{aligned}
$$

Substituting back $u = \tan(x/2)$:

$$
\int \csc x dx = \ln|\tan(x/2)| + C
$$

Question 12:

To evaluate $\int \cot x dx$, we use the substitution $x = \tan(\theta/2)$.

Then $\cot x = \frac{\cos x}{\sin x} =$
rac1-u^{2}1 $+ u^2 \frac{}{\frac{2u}{1+u^2} = \frac{1-u^2}{2u}}$ where $u = \tan(heta/2)$. Also $dx = \frac{2du}{1+u^2}$.

$$
\begin{aligned}
\int \cot x dx &= \int \frac{1-u^2}{2u} \cdot \frac{2du}{1+u^2} \\
&= \int \frac{1-u^2}{u(1+u^2)}du
\end{aligned}
$$

Using partial fractions:

$$
\begin{aligned}
\frac{1-u^2}{u(1+u^2)} &= \frac{A}{u} + \frac{Bu+C}{1+u^2} \\
1-u^2 &= A(1+u^2) + (Bu+C)u \\
\text{Let } u = 0 \implies A &= 1 \\
\text{Let } u = 1 \implies 0 = 2(1) + (B+C) \implies B+C &= -2 \\
\text{Let } u = -1 \implies 0 = 2(1) + (-B+C)(-1) \implies B-C &= 2
\end{aligned}
$$

Solving for B and C:

$$
\begin{aligned}
(B+C) + (B-C) &= -2+2 \\
2B = 0 \implies B &= 0 \\
C &= -2
\end{aligned}
$$

So the integral becomes:

$$\begin{aligned}\int \frac{1-u^2}{u(1+u^2)}du &= \int (\frac{1}{u} - \frac{2}{1+u^2})du \\ &= \ln|u| - 2\tan^{-1} u + C\end{aligned}$$

Substituting back $u = \tan(x/2)$:

$$\begin{aligned}\int \cot x dx &= \ln|\tan(x/2)| - 2\tan^{-1}(\tan(x/2)) + C \\ &= \ln|\tan(x/2)| - x + C\end{aligned}$$

Question 13:
To evaluate $\int_0^{\pi/2} \frac{1}{6+7\sin\theta} d\theta$, we use the substitution $x = \tan(\theta/2)$.

Then $\sin\theta = \frac{2x}{1+x^2}$ and $d\theta = \frac{2dx}{1+x^2}$. Limits: $\theta = 0 \implies x = 0$; $\theta = \pi/2 \implies x = 1$.

$$\begin{aligned}\int_0^{\pi/2} \frac{1}{6+7\sin\theta} d\theta &= \int_0^1 \frac{1}{6+7(\frac{2x}{1+x^2})} \cdot \frac{2dx}{1+x^2} \\ &= \int_0^1 \frac{1}{\frac{6(1+x^2)+14x}{1+x^2}} \cdot \frac{2dx}{1+x^2} \\ &= \int_0^1 \frac{2}{6x^2+14x+6} dx \\ &= \int_0^1 \frac{1}{3x^2+7x+3} dx\end{aligned}$$

Complete the square for the denominator:

$$\begin{aligned}3x^2+7x+3 &= 3(x^2 + \frac{7}{3}x + 1) \\ &= 3((x+\frac{7}{6})^2 + 1 - \frac{49}{36}) \\ &= 3((x+\frac{7}{6})^2 - \frac{13}{36})\end{aligned}$$

Now the integral is in the form $\int \frac{1}{a^2-u^2} du = \frac{1}{2a} \ln|\frac{a+u}{a-u}| + C$.

$$\int_0^1 \frac{1}{3((x+\frac{7}{6})^2 - \frac{13}{36})} dx = \frac{1}{3}\int_0^1 \frac{1}{(\frac{\sqrt{13}}{6})^2 - (x+\frac{7}{6})^2} dx$$

Wait, the form is $\displaystyle\int \frac{1}{u^2-a^2} du = \frac{1}{2a} \ln|\frac{u-a}{u+a}| + C$.

Let's re-evaluate: $3x^2+7x+3 = 3((x+\frac{7}{6})^2 - \frac{13}{36})$.

The integral is $\displaystyle\frac{1}{3}\int_0^1 \frac{1}{(x+\frac{7}{6})^2 - (\frac{\sqrt{13}}{6})^2} dx$.

Using the formula $\int \frac{1}{u^2-a^2} du = \frac{1}{2a} \ln|\frac{u-a}{u+a}| + C$:

$$
\begin{aligned}
&= \frac{1}{3}[\frac{1}{2(\frac{\sqrt{13}}{6})}\ln|\frac{x+\frac{7}{6}-\frac{\sqrt{13}}{6}}{x+\frac{7}{6}+\frac{\sqrt{13}}{6}}|]_0^1 \\
&= \frac{1}{3}[\frac{3}{\sqrt{13}}\ln|\frac{6x+7-\sqrt{13}}{6x+7+\sqrt{13}}|]_0^1 \\
&= \frac{1}{\sqrt{13}}[\ln|\frac{13-\sqrt{13}}{13+\sqrt{13}}| - \ln|\frac{7-\sqrt{13}}{7+\sqrt{13}}|]
\end{aligned}
$$

Easier Exam Questions

Question 1:

We are evaluating $\int_a^b x^9 e^{3x^{10}} dx$.

Let $u = 3x^{10}$.

Then, differentiating with respect to x:

$$
\begin{aligned}
\frac{du}{dx} &= 30x^9 \\
du &= 30x^9 dx \implies x^9 dx = \frac{1}{30} du
\end{aligned}
$$

Now we change the limits of integration:

$$
\begin{aligned}
\text{Lower limit: } x = a &\implies u = 3a^{10} \\
\text{Upper limit: } x = b &\implies u = 3b^{10}
\end{aligned}
$$

Substituting these into the integral:

$$
\begin{aligned}
\int_a^b x^9 e^{3x^{10}} dx &= \int_{3a^{10}}^{3b^{10}} e^u \left(\frac{1}{30} du\right) \\
&= \frac{1}{30}\int_{3a^{10}}^{3b^{10}} e^u du
\end{aligned}
$$

Comparing this to the options, this matches Option D.

Question 2a:

To show that $\sin(A+B)+\sin(A-B) = 2\sin A\cos B$, we use the angle addition formulas for sine:

$$
\begin{aligned}
\sin(A+B) &= \sin A\cos B + \cos A\sin B \\
\sin(A-B) &= \sin A\cos B - \cos A\sin B
\end{aligned}
$$

Adding these two equations together:

$$
\begin{aligned}
\sin(A+B) + \sin(A-B) &= (\sin A\cos B + \cos A\sin B) + (\sin A\cos B - \cos A\sin B) \\
&= 2\sin A\cos B
\end{aligned}
$$

Question 2b:

We need to evaluate $\int_0^{\pi/2} \sin 9x \cos 6x dx$.

Using the identity from part (a), let $A = 9x$ and $B = 6x$. Then $A+B = 15x$ and $A-B = 3x$. The identity $2\sin A\cos B = \sin(A+B) + \sin(A-B)$ implies:

$$\sin 9x\cos 6x = \frac{1}{2}(\sin 15x + \sin 3x)$$

Now we integrate:

$$\begin{aligned}
\int_0^{\pi/2} \sin 9x\cos 6x dx &= \frac{1}{2}\int_0^{\pi/2}(\sin 15x + \sin 3x)dx \\
&= \frac{1}{2}[-\frac{1}{15}\cos 15x - \frac{1}{3}\cos 3x]_0^{\pi/2} \\
&= -\frac{1}{2}[\frac{1}{15}\cos(15\pi/2) + \frac{1}{3}\cos(3\pi/2) - (\frac{1}{15}\cos 0 + \frac{1}{3}\cos 0)] \\
&= -\frac{1}{2}[0 + 0 - (\frac{1}{15} + \frac{1}{3})] \\
&= \frac{1}{2}(\frac{1+5}{15}) = \frac{1}{2}\cdot\frac{6}{15} = \frac{3}{15} = \frac{1}{5}
\end{aligned}$$

Question 3:
To evaluate $\int \sin^3 x\cos^2 x dx$, we use the identity $\sin^2 x = 1 - \cos^2 x$ to express the odd power of sine in terms of cosine.

$$\begin{aligned}
\int \sin^3 x\cos^2 x dx &= \int (\sin^2 x)\cos^2 x(\sin x dx) \\
&= \int (1-\cos^2 x)\cos^2 x(\sin x dx)
\end{aligned}$$

Let $u = \cos x$, then $du = -\sin x dx$, so $\sin x dx = -du$.

$$\begin{aligned}
\int (1-u^2)u^2(-du) &= \int (u^4 - u^2)du \\
&= \frac{u^5}{5} - \frac{u^3}{3} + C
\end{aligned}$$

Substituting back $u = \cos x$:

$$\int \sin^3 x\cos^2 x dx = \frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + C$$

Question 4c:
To evaluate $\int_{3\sqrt{2}}^{6} \frac{1}{x^2\sqrt{x^2-9}}dx$, we use the substitution $x = 3\sec\theta$.

Then $dx = 3\sec\theta\tan\theta d\theta$ and $\sqrt{x^2-9} = \sqrt{9\sec^2\theta - 9} = 3\tan\theta$.

Limits: $x = 3\sqrt{2} \implies 3\sec\theta = 3\sqrt{2} \implies \sec\theta = \sqrt{2} \implies \theta = \pi/4$. $x = 6 \implies 3\sec\theta = 6 \implies \sec\theta = 2 \implies \theta = /3$.

$$\begin{aligned}
\int_{3\sqrt{2}}^{6} \frac{1}{x^2\sqrt{x^2-9}}dx &= \int_{\pi/4}^{\pi/3} \frac{3\sec\theta\tan\theta d\theta}{(3\sec\theta)^2(3\tan\theta)} \\
&= \int_{\pi/4}^{\pi/3} \frac{3\sec\theta\tan\theta}{9\sec^2\theta\cdot 3\tan\theta}d\theta \\
&= \int_{\pi/4}^{\pi/3} \frac{1}{9\sec\theta}d\theta \\
&= \frac{1}{9}\int_{\pi/4}^{\pi/3} \cos\theta d\theta \\
&= \frac{1}{9}[\sin\theta]_{\pi/4}^{\pi/3} \\
&= \frac{1}{9}(\sin(\pi/3)-\sin(\pi/4)) \\
&= \frac{1}{9}(\frac{\sqrt{3}}{2}-\frac{\sqrt{2}}{2}) = \frac{\sqrt{3}-\sqrt{2}}{18}
\end{aligned}$$

Question 4d:
To find $\int \cos^5 x dx$, we split the power of cosine into an even power and a single cosine term.

$$\begin{aligned}
\int \cos^5 x dx &= \int (\cos^2 x)^2 \cos x dx \\
&= \int (1-\sin^2 x)^2 \cos x dx
\end{aligned}$$

Let $u = \sin x$, then $du = \cos x dx$.

$$\begin{aligned}
\int (1-u^2)^2 du &= \int (1-2u^2+u^4)du \\
&= u - \frac{2u^3}{3} + \frac{u^5}{5} + C
\end{aligned}$$

Substituting back $u = \sin x$:

$$\int \cos^5 x dx = \sin x - \frac{2}{3}\sin^3 x + \frac{1}{5}\sin^5 x + C$$

Question 4e:
To find $\int \frac{1}{1+\cos x}dx$, we can use the half-angle identity $\cos x = 2\cos^2(x/2) - 1$.

$$\begin{aligned}
\int \frac{1}{1+\cos x}dx &= \int \frac{1}{1+(2\cos^2(x/2)-1)}dx \\
&= \int \frac{1}{2\cos^2(x/2)}dx \\
&= \frac{1}{2}\int \sec^2(x/2)dx
\end{aligned}$$

Integrating $\sec^2(x/2)$:

$$\int \sec^2(x/2)dx = \frac{1}{1/2}\tan(x/2) = 2\tan(x/2)$$

So:

$$\int \frac{1}{1+\cos x}dx = \frac{1}{2}(2\tan(x/2)) + C = \tan(x/2) + C$$

Question 5:
We are evaluating $\int_0^{\sqrt{3}} \frac{\ln(\tan^{-1} x)}{1+x^2}dx$.
Let $u = \tan^{-1} x$. Then $du = \frac{1}{1+x^2}dx$.
Limits: $x = 0 \implies u = \tan^{-1} 0 = 0$. $x = \sqrt{3} \implies u = \tan^{-1}\sqrt{3} = \pi/3$.

$$\int_0^{\sqrt{3}} \frac{\ln(\tan^{-1} x)}{1+x^2}dx = \int_0^{\pi/3} \ln u\, du$$

Comparing this to the options, this matches Option A.

Question 6i:
We are evaluating $I = \int_1^{\sqrt{3}} \frac{dx}{x(1+x^2)}$.
Let $x = \tan u$. Then $dx = \sec^2 u du$ and $1 + x^2 = \sec^2 u$.
Limits: $x = 1 \implies \tan u = 1 \implies u = /4$. $x = \sqrt{3} \implies \tan u = \sqrt{3} \implies u = /3$.

$$\begin{aligned} I &= \int_{\pi/4}^{\pi/3} \frac{\sec^2 u du}{\tan u(\sec^2 u)} \\ &= \int_{\pi/4}^{\pi/3} \frac{du}{\tan u} \end{aligned}$$

Thus, $a = /4$ and $b = /3$.

Question 6ii:
From part (i), $I = \int_{\pi/4}^{\pi/3} \frac{du}{\tan u} = \int_{\pi/4}^{\pi/3} \cot u du$.
Integrating $\cot u$:

$$\begin{aligned} I &= [\ln|\sin u|]_{\pi/4}^{\pi/3} \\ &= \ln|\sin(\pi/3)| - \ln|\sin(\pi/4)| \\ &= \ln(\frac{\sqrt{3}}{2}) - \ln(\frac{\sqrt{2}}{2}) \\ &= \ln\left(\frac{\sqrt{3}/2}{\sqrt{2}/2}\right) = \ln(\frac{\sqrt{3}}{\sqrt{2}}) \\ &= \ln(\sqrt{3/2}) = \frac{1}{2}\ln(3/2) \\ &= \frac{1}{2}(\ln 3 - \ln 2) \end{aligned}$$

Question 7:
To evaluate $\int_0^1 x^2\sqrt{1-x^2}dx$, we use the substitution $x = \sin\theta$.
Then $dx = \cos\theta d\theta$ and $\sqrt{1-x^2} = \cos\theta$.
Limits: $x = 0 \implies \theta = 0$; $x = 1 \implies \theta = /2$.

$$
\begin{aligned}
\int_0^1 x^2\sqrt{1-x^2}dx &= \int_0^{\pi/2} (\sin^2\theta)(\cos\theta)(\cos\theta d\theta) \\
&= \int_0^{\pi/2} \sin^2\theta\cos^2\theta d\theta \\
&= \int_0^{\pi/2} (\sin\theta\cos\theta)^2 d\theta \\
&= \int_0^{\pi/2} (\frac{1}{2}\sin 2\theta)^2 d\theta \\
&= \frac{1}{4}\int_0^{\pi/2} \sin^2 2\theta d\theta \\
&= \frac{1}{4}\int_0^{\pi/2} \frac{1-\cos 4\theta}{2} d\theta \\
&= \frac{1}{8}[\theta - \frac{1}{4}\sin 4\theta]_0^{\pi/2} \\
&= \frac{1}{8}((\pi/2-0)-(0-0)) = \frac{\pi}{16}
\end{aligned}
$$

Harder Exam Questions

Question 1:

We use the substitution $t = \tan(x/2)$.

Then $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, and $dx = \frac{2dt}{1+t^2}$.

Limits: $x = 0 \implies t = 0$; $x = \pi/2 \implies t = 1$.

$$
\begin{aligned}
\int_0^{\pi/2} \frac{a}{5+3\sin x+4\cos x} dx &= \int_0^1 \frac{a}{5+3(\frac{2t}{1+t^2})+4(\frac{1-t^2}{1+t^2})} \cdot \frac{2dt}{1+t^2} \\
&= \int_0^1 \frac{2a}{5(1+t^2)+6t+4(1-t^2)} dt \\
&= \int_0^1 \frac{2a}{5+5t^2+6t+4-4t^2} dt \\
&= \int_0^1 \frac{2a}{t^2+6t+9} dt \\
&= \int_0^1 \frac{2a}{(t+3)^2} dt
\end{aligned}
$$

Integrating:

$$
\begin{aligned}
\int_0^1 2a(t+3)^{-2} dt &= [-\frac{2a}{t+3}]_0^1 \\
&= -\frac{2a}{4} - (-\frac{2a}{3}) \\
&= -\frac{a}{2} + \frac{2a}{3} \\
&= \frac{-3a+4a}{6} = \frac{a}{6}
\end{aligned}
$$

Given that the integral equals 1:

$$
\frac{a}{6} = 1 \implies a = 6
$$

Question 2:
We evaluate $\int_{1/2}^{1} \sqrt{\frac{1-u}{1+u}} du$ using the substitution $u = \cos 2\theta$.

Then $du = -2\sin 2\theta d\theta$.

Limits:
$u = 1/2 \implies \cos 2\theta = 1/2 \implies 2\theta = \pi/3 \implies \theta = \pi/6.$
$u = 1 \implies \cos 2\theta = 1 \implies 2\theta = 0 \implies \theta = 0.$

$$\sqrt{\frac{1-u}{1+u}} = \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} = \sqrt{\frac{2\sin^2\theta}{2\cos^2\theta}} = \sqrt{\tan^2\theta} = \tan\theta$$

Substituting into the integral:

$$\begin{aligned}
\int_{1/2}^{1} \sqrt{\frac{1-u}{1+u}} du &= \int_{\pi/6}^{0} \tan\theta(-2\sin 2\theta d\theta) \\
&= -2\int_{\pi/6}^{0} \tan\theta(2\sin\theta\cos\theta) d\theta \\
&= -4\int_{\pi/6}^{0} \frac{\sin\theta}{\cos\theta} \sin\theta\cos\theta d\theta \\
&= -4\int_{\pi/6}^{0} \sin^2\theta d\theta \\
&= 4\int_{0}^{\pi/6} \sin^2\theta d\theta \\
&= 4\int_{0}^{\pi/6} \frac{1-\cos 2\theta}{2} d\theta \\
&= 2\int_{0}^{\pi/6} (1-\cos 2\theta) d\theta \\
&= 2[\theta - \frac{1}{2}\sin 2\theta]_0^{\pi/6} \\
&= 2((\pi/6 - \frac{1}{2}\sin(\pi/3)) - (0-0)) \\
&= 2(\pi/6 - \frac{1}{2}\cdot\frac{\sqrt{3}}{2}) \\
&= \frac{\pi}{3} - \frac{\sqrt{3}}{2}
\end{aligned}$$

Question 3c:
To find $\int \frac{1}{1+e^{-x}} dx$, we multiply the numerator and denominator by e^x:

$$\begin{aligned}
\int \frac{1}{1+e^{-x}} dx &= \int \frac{e^x}{e^x(1+e^{-x})} dx \\
&= \int \frac{e^x}{e^x+1} dx
\end{aligned}$$

Let $u = e^x + 1$, then $du = e^x dx$.

$$\begin{aligned}
\int \frac{1}{u} du &= \ln|u| + C \\
&= \ln|e^x + 1| + C
\end{aligned}$$

Question 3d:
To find $\int \frac{1}{(1+x^2)^{3/2}}dx$, we use the substitution $x = \tan\theta$.

Then $dx = \sec^2\theta d\theta$ and $1 + x^2 = \sec^2\theta$.

$$\begin{aligned}\int \frac{1}{(1+x^2)^{3/2}}dx &= \int \frac{\sec^2\theta d\theta}{(\sec^2\theta)^{3/2}} \\ &= \int \frac{\sec^2\theta}{\sec^3\theta}d\theta \\ &= \int \frac{1}{\sec\theta}d\theta \\ &= \int \cos\theta d\theta \\ &= \sin\theta + C\end{aligned}$$

From $x = \tan\theta$, we have $\sin\theta = \frac{x}{\sqrt{1+x^2}}$.

$$\int \frac{1}{(1+x^2)^{3/2}}dx = \frac{x}{\sqrt{1+x^2}} + C$$

Question 4a:
To evaluate $\int_0^1 \frac{x+3}{\sqrt{4-2x-x^2}}dx$, we first complete the square for the quadratic in the square root:

$$\begin{aligned}4 - 2x - x^2 &= -(x^2 + 2x - 4) \\ &= -((x+1)^2 - 5) \\ &= 5 - (x+1)^2\end{aligned}$$

Now we rewrite the integral:

$$\int_0^1 \frac{x+3}{\sqrt{5-(x+1)^2}}dx$$

Let $u = x + 1$, then $du = dx$ and $x = u - 1$. The limits become $x = 0 \implies u = 1$ and $x = 1 \implies u = 2$.

$$\begin{aligned}\int_1^2 \frac{(u-1)+3}{\sqrt{5-u^2}}du &= \int_1^2 \frac{u+2}{\sqrt{5-u^2}}du \\ &= \int_1^2 \frac{u}{\sqrt{5-u^2}}du + \int_1^2 \frac{2}{\sqrt{5-u^2}}du\end{aligned}$$

For the first part, let $v = 5 - u^2$, then $dv = -2udu$, so $udu = -1/2dv$.

$$\begin{aligned}\int_1^2 \frac{u}{\sqrt{5-u^2}}du &= -\frac{1}{2}\int_{5-1^2}^{5-2^2} v^{-1/2}dv \\ &= -\frac{1}{2}[2v^{1/2}]_4^1 \\ &= -[\sqrt{v}]_4^1 = -(\sqrt{1}-\sqrt{4}) = -(1-2) = 1\end{aligned}$$

For the second part, we use the formula $\int \frac{1}{\sqrt{a^2-u^2}}du = \sin^{-1}(u/a)$.

$$\int_1^2 \frac{2}{\sqrt{5-u^2}}du = 2[\sin^{-1}(u/\sqrt{5})]_1^2$$
$$= 2(\sin^{-1}(2/\sqrt{5}) - \sin^{-1}(1/\sqrt{5}))$$

Summing the results:

$$\int_0^1 \frac{x+3}{\sqrt{4-2x-x^2}}dx = 1 + 2(\sin^{-1}(2/\sqrt{5}) - \sin^{-1}(1/\sqrt{5}))$$

Question 4b:
To find $\int \sin^4 x \cos^5 x dx$, we use the identity $\cos^2 x = 1 - \sin^2 x$ to express the odd power of cosine in terms of sine.

$$\int \sin^4 x \cos^5 x dx = \int \sin^4 x (\cos^2 x)^2 \cos x dx$$
$$= \int \sin^4 x (1 - \sin^2 x)^2 \cos x dx$$

Let $u = \sin x$, then $du = \cos x dx$.

$$\int u^4 (1-u^2)^2 du = \int u^4 (1 - 2u^2 + u^4) du$$
$$= \int (u^4 - 2u^6 + u^8) du$$
$$= \frac{u^5}{5} - \frac{2u^7}{7} + \frac{u^9}{9} + C$$

Substituting back $u = \sin x$:

$$\int \sin^4 x \cos^5 x dx = \frac{1}{5}\sin^5 x - \frac{2}{7}\sin^7 x + \frac{1}{9}\sin^9 x + C \tag{1}$$

Question 5:
To find $\int \frac{\tan^2 x + 2}{x + \tan x} dx$, we observe the relationship between the numerator and the denominator.
Let $u = x + \tan x$.
Then, differentiating with respect to x:

$$\frac{du}{dx} = 1 + \sec^2 x$$
$$= 1 + (1 + \tan^2 x)$$
$$= 2 + \tan^2 x$$

We see that the numerator is exactly the derivative of the denominator.

$$\int \frac{\tan^2 x + 2}{x + \tan x} dx = \int \frac{1}{u} du$$
$$= \ln|u| + C$$
$$= \ln|x + \tan x| + C$$